

CH4203: Organic Synthesis Laboratory

Expt. No. 6: Copper-Catalyzed Click Chemistry of an Enantioenriched Azide with a Terminal Alkyne

Quiz

Marks: 5 (Each Question would carry 1.0 Mark)

(Q1) The mechanism of Copper-Catalyzed Click Chemistry could be considered as:

- (a) 1,3-Sigmatropic rearrangement
- (b) [3+2]-Formal Cycloaddition
- (c) Substitution Reaction
- (d) Cross-Coupling Pathway

Correct Answer: (b)

(Q2) CuSO_4 can be reacted with Sodium Ascorbate to generate Cu(I)-species. This reaction is:

- (a) Oxidation of Copper center
- (b) Oxidation of Sodium Ascorbate
- (c) Reduction of Sodium Ascorbate
- (d) None of these

Correct Answer: (b)

(Q3) Conversion of N-Boc L-prolinol to N-Boc L-prolinyl azide using PPh_3 and DIAD (Dilsopropyl AzoDicarboxylate) follows a:

- (a) Stetter Reaction
- (b) Mitsunobu Reaction
- (c) Elimination Reaction
- (d) Hoffman Degradation

Correct Answer: (b)

(Q4) Disubstituted acetylenes can be converted to Z-selective olefin under:

- (a) Clemmensen Reduction
- (b) LiAlH_4 Reduction
- (c) Birch Reduction
- (d) Lindlar Reduction

Correct Answer: (d)

(Q5) The trajectory for nucleophilic approach to a terminal alkyne would follow:

- (a) 120 degree angle
- (b) 107 degree angle
- (c) 180 degree angle
- (d) 90 degree angle

Correct Answer: (a)

(Q6) The Copper-Catalyzed Click Chemistry of azide and terminal alkyne is highly regioselective in nature and produced:

- (a) 1,4-disubstituted 1,2,3-triazole
- (b) 1,2-disubstituted 1,2,4-triazole
- (c) 1,3-disubstituted 1,2,3-triazole
- (d) 1,4-disubstituted 1,2,4-triazole

Correct Answer: (a)

(Q7) The Copper-Catalyzed Click Chemistry of azide and terminal alkyne involve:

- (a) Cu(I) acetylide
- (b) Cu(II) acetylide
- (c) Zn(I) acetylide
- (d) none of these

Correct Answer: (a)

(Q8) Along with an organic azide, what other important reactant is always needed for a Huisgen cycloaddition?

- (a) an alkyne
- (b) an Alkene
- (c) an Aldehyde
- (d) a Ketone

Correct Answer: (a)

(Q9) In ozonolysis of an unsymmetrically disubstituted olefin produce 2 different types of ketones. How many formal [3+2]-cycloaddition involve during ozonolysis process?

- (a) 2
- (b) 3
- (c) 4
- (d) 1

Correct Answer: (a)

(Q10) Staudinger reaction of an azide with triphenylphosphine in water produce:

- (a) an amine and triphenylphosphine oxide
- (b) an imine and triphenylphosphine oxide
- (c) an aldehyde and triphenylphosphine oxide
- (d) an amine and triphenylphosphine

Correct Answer: (a)